

Geometric Distribution

First success in n trials $\rightarrow$ Sequence of independent Bernoulli events with probability p

$\Rightarrow$ What is the probability that the first success (first $B_n=1$) occurs on the n^{th} trial?

$$\begin{aligned} P(\text{first success on } n^{\text{th}}) &= P(F_1 \cap F_2 \cap \dots \cap F_{n-1} \cap S_n) = \\ &= P(F_1) P(F_2) \dots P(F_{n-1}) P(S_n) = \\ &= (1-p)^{n-1} p = q^{n-1} p \end{aligned}$$

$$X \sim \text{geometric}(p)$$

$$P(X=n) = q^{n-1} p$$

Ex: Prob it takes at least 10 rolls of a dice to roll a 5?

$$P(X \geq 10) = P(X \geq 10) + (P \geq 11) + (P \geq 12) + \dots =$$

$$= q^9 p + q^{10} p + q^{11} p + \dots =$$

$$= q^9 p (1 + q + q^2 + q^3 + \dots) =$$

$$= \frac{q^9 p}{1-q} = q^9 = \left(\frac{5}{6}\right)^9$$

$q < 1 \Rightarrow$ Geometric Sequence

Another way: $X \sim \text{geometric}(1/6)$